Advanced Algebra

ANSWER KEY

Simplifying rational expressions

1 Simplify $\frac{x^2-4}{x^2+x-6}$.

$$\frac{x^2-4}{x^2+x-6}=\frac{(x-2)(x+2)}{(x-2)(x+3)}=\frac{x+2}{x+3},\ \ x\neq 2.$$

2 Simplify $\frac{x^2-5x+6}{x^2-9}$.

$$\frac{x^2-5x+6}{x^2-9}=\frac{(x-2)(x-3)}{(x-3)(x+3)}=\frac{x-2}{x+3},\ \ x\neq 3.$$

3 Combine into a single fraction: $\frac{2}{x+1}+\frac{3}{x-1}$.

$$\frac{2}{x+1}+\frac{3}{x-1}=\frac{2(x-1)+3(x+1)}{(x+1)(x-1)}=\frac{2x-2+3x+3}{x^2-1}=\frac{5x+1}{x^2-1}.$$

4 Simplify $\frac{3x^2-12}{x^2-4x+4}$.

$$\frac{3x^2-12}{x^2-4x+4}=\frac{3(x-2)(x+2)}{(x-2)^2}=\frac{3(x+2)}{x-2},\ \ x\neq 2.$$

Absolute value equations

5 Solve $|2x-5|=9$.

$2x-5=9\Rightarrow x=7$. $2x-5=-9\Rightarrow x=-2$. Solutions: $x=7$ or $x=-2$.

6 Solve $|3x+1|=|x-5|$.

Case 1: $3x+1=x-5\Rightarrow 2x=-6\Rightarrow x=-3$. Case 2: $3x+1=-(x-5)\Rightarrow 4x=4\Rightarrow x=1$. Both check: solutions $x=-3$ or $x=1$.

7 Solve $|x-4|+3=10$.

$|x-4|=7$, so $x-4=7\Rightarrow x=11$, or $x-4=-7\Rightarrow x=-3$. Solutions: $x=11$ or $x=-3$.

8 Solve $|4x+3|=11$.

$4x+3=11\Rightarrow x=2$. $4x+3=-11\Rightarrow x=-3.5$. Solutions: $x=2$ or $x=-3.5$.

Polynomial division, remainder & factor theorems

9 Use the remainder theorem to find the remainder when $f(x)=x^3+2x^2-5x+1$ is divided by $(x-2)$.

$f(2)=8+8-10+1=7$. The remainder is 7.

10 Given $f(x)=x^3+3x^2-4x-12$:

a Use the factor theorem to show that $(x+3)$ is a factor of $f(x)$.
$f(-3)=(-27)+3(9)-4(-3)-12=-27+27+12-12=0$, so by the factor theorem $(x+3)$ is a factor.

b Hence factor $f(x)$ completely. Dividing: $f(x)=(x+3)(x^2-4)=(x+3)(x-2)(x+2)$.

11 Divide $x^3 - 2x^2 - 5x + 6$ by $(x - 3)$ using polynomial long division. State the quotient and remainder.
Quotient $= x^2 + x - 2$, remainder $= 0$. So
$x^3 - 2x^2 - 5x + 6 = (x - 3)(x^2 + x - 2) = (x - 3)(x + 2)(x - 1)$.

Logarithmic equations

12 Solve $\log_3 (x + 5) = 2$.
$x + 5 = 3^2 = 9 \Rightarrow x = 4$.

13 Solve $\log_2 (x + 1) + \log_2 (x - 1) = 3$.
$\log_2 [(x + 1)(x - 1)] = 3 \Rightarrow x^2 - 1 = 8 \Rightarrow x^2 = 9 \Rightarrow x = \pm 3$. Since $x > 1$ is required for $\log_2 (x - 1)$ to be defined, reject $x = -3$. Solution: $x = 3$.

14 Solve $2\log_5 x = \log_5 16$.
$\log_5 (x^2) = \log_5 16 \Rightarrow x^2 = 16 \Rightarrow x = \pm 4$. Since $x > 0$ is required, reject $x = -4$. Solution: $x = 4$.

15 Solve $\log_3 (x - 2) = 4$.
$x - 2 = 3^4 = 81 \Rightarrow x = 83$.

Exponential equations

16 Solve $3^{2x-1} = 27$.
$27 = 3^3$, so $2x - 1 = 3 \Rightarrow x = 2$.

17 Solve $5^{x+2} = 5^{3x-4}$.
$x + 2 = 3x - 4 \Rightarrow 6 = 2x \Rightarrow x = 3$.

18 Solve $2^x = 15$ for x, giving your answer to three decimal places.
$x = \dfrac{\ln 15}{\ln 2} \approx \dfrac{2.708}{0.693} \approx 3.907$.

19 Solve $4^x = 2^{x+3}$.
Rewrite $4^x = 2^{2x}$, so $2x = x + 3 \Rightarrow x = 3$.

Nonlinear systems of equations

20 Solve the system $y = x^2 - 1$ and $y = 2x + 2$.
$x^2 - 1 = 2x + 2 \Rightarrow x^2 - 2x - 3 = 0 \Rightarrow (x - 3)(x + 1) = 0 \Rightarrow x = 3$ or $x = -1$. Solutions: $(3, 8)$ and $(-1, 0)$.

21 Solve the system $x^2 + y^2 = 25$ and $y = x + 1$.

Substitute: $x^2 + (x + 1)^2 = 25 \Rightarrow 2x^2 + 2x - 24 = 0 \Rightarrow x^2 + x - 12 = 0 \Rightarrow (x + 4)(x - 3) = 0 \Rightarrow x = -4$ or $x = 3$. Solutions: $(-4, -3)$ and $(3, 4)$.

22 Solve the system $y = x^2 - 4x + 3$ and $y = -x + 3$.

$x^2 - 4x + 3 = -x + 3 \Rightarrow x^2 - 3x = 0 \Rightarrow x(x - 3) = 0 \Rightarrow x = 0$ or $x = 3$. Solutions: $(0, 3)$ and $(3, 0)$.

Rational equations and extraneous solutions

23 Solve $\frac{4}{x - 1} = \frac{2}{x + 1}$.

Cross-multiplying: $4(x + 1) = 2(x - 1) \Rightarrow 4x + 4 = 2x - 2 \Rightarrow 2x = -6 \Rightarrow x = -3$. Valid, since $x \neq \pm 1$.

24 Solve $\frac{x}{x - 2} + \frac{2}{x + 2} = \frac{8}{x^2 - 4}$.

Multiply through by $(x - 2)(x + 2)$:

$x(x + 2) + 2(x - 2) = 8 \Rightarrow x^2 + 4x - 4 = 8 \Rightarrow x^2 + 4x - 12 = 0 \Rightarrow (x + 6)(x - 2) = 0 \Rightarrow x = -6$ or $x = 2$. Since $x = 2$ makes the denominator zero, it is extraneous. Solution: $x = -6$.

25 Solve $\frac{3}{x + 4} = \frac{1}{x - 2}$.

Cross-multiplying: $3(x - 2) = x + 4 \Rightarrow 3x - 6 = x + 4 \Rightarrow 2x = 10 \Rightarrow x = 5$. Valid, since $x \neq -4, 2$.

Systems of inequalities and linear programming

26 A feasible region is bounded by the constraints $x \geq 0$, $y \geq 0$, $x + y \leq 6$, and $2x + y \leq 10$. Find the vertices of the feasible region.

Intersecting boundary lines pairwise (and checking each candidate point satisfies every constraint) gives vertices $(0, 0)$, $(5, 0)$, $(4, 2)$, and $(0, 6)$.

27 Using the feasible region from the previous question, find the maximum value of $P = 3x + 2y$ over the region, and state where it occurs.

Evaluating P at each vertex: $(0, 0) \rightarrow 0$; $(5, 0) \rightarrow 15$; $(4, 2) \rightarrow 16$; $(0, 6) \rightarrow 12$. The maximum is $P = 16$, occurring at $(4, 2)$.

A well-designed word problem: production planning at a furniture workshop

28 This question has three parts. A furniture workshop makes tables and chairs. Each table requires 4 hours of carpentry and 2 hours of finishing; each chair requires 1 hour of carpentry and 1 hour of finishing. Each week the shop has at most 32 hours of carpentry time and 20 hours of finishing time available. Each table earns a profit of 70 dollars and each chair earns a profit of 30 dollars. Let x be the number of tables and y the number of chairs produced in a week. (a) Write the system of inequalities describing the constraints (including $x \geq 0$ and $y \geq 0$). (b) Find the vertices of the feasible region. (c) Determine the number of tables and chairs that maximizes the weekly profit $P = 70x + 30y$, and find the maximum profit.

(a) The constraints are $x \geq 0$, $y \geq 0$, $4x + y \leq 32$ (carpentry), and $2x + y \leq 20$ (finishing). (b) The boundary lines meet the axes and each other at $(0, 0)$, $(8, 0)$, $(6, 8)$, and $(0, 20)$ — these are the vertices of the feasible region. (c) Evaluating $P = 70x + 30y$: $(0, 0) \rightarrow 0$; $(8, 0) \rightarrow 560$; $(6, 8) \rightarrow 660$; $(0, 20) \rightarrow 600$. The maximum profit is 660 dollars, achieved by producing 6 tables and 8 chairs per week.